# Muhammad Islam Kamran

**Senior Backend & Distributed Systems Engineer**

High-Throughput APIs · Microservices (SOA) · Event-Driven Systems · LLM/AI Pipelines · PostgreSQL

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## Technical Skills

**Languages & Frameworks:** Python (FastAPI, Django, Flask, AsyncIO), Service-Oriented Architecture (SOA), RESTful APIs, gRPC, WebSockets, TypeScript, SQL, Bash, Clean Code (PEP 8)
**Distributed Systems & Streaming:** Apache Kafka (FIFO Partitioning, Event-Driven Architecture), Redis (Cache-Aside, Rate Limiting), Celery, RabbitMQ, PostgreSQL, MySQL, Pandas, DuckDB
**Financial & Data Integrity:** 3-Way ACID Reconciliation, Audit Trails, Idempotency, Payment Webhook Pipelines, pgvector Embeddings, LLM Integration (OpenAI Whisper, Prompt Pipelines)
**Cloud, Infra & Testing:** AWS (EC2, S3, RDS, CloudWatch), Docker, Kubernetes (EKS), Nginx, Linux Administration, CI/CD (GitHub Actions), Zero-Downtime Rollouts, PyTest (>85% Coverage)
**Engineering Leadership:** Squad Leadership (4 Engineers), Mandatory Peer Code Reviews, Shift-Left Testing, Incident Response, High Availability (99.95% SLA)

## Professional Experience

**Indus Real Estate LLC – Apparel Group** **March 2025 – Present**
*Team Lead – Software Engineering & Architecture* *Dubai, UAE*

- Architect and maintain high-throughput backend microservices using **Python/FastAPI** and **PostgreSQL**, optimizing database queries and Redis cache-aside layers to slash latency from **1.8s to <220ms (87% reduction)**.
- Engineered a 3-way reconciliation and financial settlement engine featuring **ACID compliance**, idempotent retry logic, and distributed audit logging, reducing transaction discrepancies by **99.4%**.
- Designed event-driven streaming layers utilizing **Apache Kafka** with per-entity partition-key ordering (<200ms latency) and **WebSockets** for real-time inventory updates across enterprise client portals.
- Built intelligent inquiry classification pipelines integrating **LLM APIs**, parsing unstructured customer requests and routing verified leads with automated intent scoring and CRM sync.
- Direct an agile squad of 4 engineers, enforcing mandatory peer code reviews, definition-of-done criteria, and automated **PyTest** suites maintaining **>85% test coverage** across all CI/CD builds.

**Business Innovation & Tech Solutions (BITS)** **May 2024 – Present**
*Lead Backend Engineer & Technical Consultant* *Hybrid – Dubai / Pakistan*

- Designed scalable microservices in **Python (FastAPI)** and **TypeScript** for enterprise automotive trading platforms, handling high-concurrency order validation and third-party webhook processing.
- Implemented sliding-window rate limiters and multi-tier caching via **Redis**, safeguarding critical public endpoints against burst traffic and ensuring **99.95% platform availability**.
- Engineered asynchronous background task pipelines using **AsyncIO** and worker queues, processing bulk inventory reconciliation and catalog synchronization with zero data loss.
- Orchestrated containerized deployments using **Docker** and **AWS**, establishing automated GitHub Actions pipelines with zero-downtime rolling deployment strategies.

**CareCloud** **October 2022 – April 2023**
*Python Backend Engineer* *United States (Remote)*

- Built scalable backend API services in **Django** for Rosetta Stone Language Lab, integrating **OpenAI Whisper speech-to-text models** for automated real-time student pronunciation scoring.
- Engineered intelligent medical document extraction pipelines (OCR & Bots) in **Flask**, parsing structured clinical and billing data from 47+ distinct US health insurance card schemas.
- Developed diagnostic data ingestion pipelines for the FOX Caregiver platform across 21 US states, converting unstructured physician notes into structured records in **MongoDB**.

## Research Publication & Key Systems

**Quantum Walk Image Encryption Scheme** | *MATLAB, Chaos Theory, Cryptography* **ResearchGate**

- First-author of peer-reviewed paper published in **CMC (Impact Factor 3.7)**, designing secure confusion-diffusion encryption algorithms using Quantum Walk and chaotic logistic maps. Paper Link

**High-Volume Ingestion & Document Processing Pipeline** | *Python, AsyncIO, S3, Docker, AWS* **2025**

- Architected an asynchronous S3 presigned-URL ingestion service with parallel worker queues, accelerating batch document processing from 15 min to 3 min with 98% schema accuracy.

## Education

**Quaid-i-Azam University (QAU)** **2020 – 2022**
*Master of Science (MS) in Computer Science* *Islamabad, Pakistan*

**University of Engineering & Technology (UET)** **2016 – 2020**
*Bachelor of Science (BS) in Computer Systems Engineering* *Peshawar, Pakistan*